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Breast Cancer Screening in the Average-Risk Patient

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Continuing Education Activity

Breast cancer is the most common non-skin cancer and the second leading cause of cancer-related deaths in women. Early detection through screening can significantly reduce both morbidity and mortality. Screening recommendations distinguish between women at average and high risk for developing **breast** cancer, as this risk stratification guides the selection of screening methods and timing. Most major guidelines recommend starting routine mammography at age 40 for women at average risk. Although the frequency of screening—either annually or biennially—varies among guidelines, the overall trend is toward increased surveillance due to its potential to save lives.

Clinical **breast** examinations are recommended every 1 to 3 years for women aged 25 to 39 and annually for women aged 40 or older. Screening typically continues until the patient's life expectancy is less than age 10, with the decision to discontinue based on individual health status and overall life expectancy. This activity reviews **breast** cancer screening for average-risk patients and offers healthcare professionals guidance on assessing **breast** cancer risk, conducting appropriate screenings, and implementing interprofessional management strategies to enhance patient outcomes. This activity also highlights the significance of multidisciplinary collaboration among healthcare providers to ensure comprehensive and personalized care for patients with varying levels of **breast** cancer risk.

Objectives:

- Identify the criteria that classify patients as average-risk or high-risk patients for **breast** cancer to ensure appropriate screening practices.
- Implement evidence-based practices for **breast** cancer screening, ensuring adherence to current guidelines and recommendations.
- Select appropriate imaging modalities, such as MRI and mammography, based on patient risk factors and preferences.
- Collaborate with an interprofessional healthcare team to enhance the overall management of **breast** cancer screening and follow-up care for patients.

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Introduction

Breast cancer is the most common malignancy in women,[1] and it is the most prevalent non-skin cancer. **Breast** cancer is also considered the second leading cause of cancer-related deaths in women. Stage IV **breast** cancer currently has no curative treatment and is managed with palliative care. Early detection of tumors significantly reduces morbidity and mortality. The primary challenge lies in determining who should be screened and when. Additionally, as our understanding of the pathophysiology of **breast** cancer develops, treatment decisions are shifting away from focusing solely on tumor size and extent. Increasingly, the incorporation of genetic and biological characteristics is being used to guide prognosis and treatment.[2]

In addition to self-**breast** examinations, the primary screening methods include clinical **breast** evaluations and imaging techniques such as mammography, ultrasonography, and magnetic resonance imaging (MRI). Multiple randomized studies have established that routine screening mammography should be offered to women aged 50 to 69. Historically, the recommendation for routine mammograms for women aged 40 to 49 or those aged 70 or older has been debated. However, the latest guidelines from the United States Preventive Services Task Force (USPSTF), the American College of Obstetricians and Gynecologists (ACOG), the American Society of **Breast** Surgeons (ASBrS), and the National Comprehensive Cancer Network (NCCN) now advocate starting routine mammograms at age 40.[3]

The frequency of screening remains a topic of debate, with some groups recommending annual screening for average-risk individuals while others advocate for biennial screening. Overall, there has been a trend toward expanded screening, as **breast** cancer detection at early and treatable stages has saved countless lives. Differentiating between average-risk and high-risk patients is crucial for optimizing patient outcomes, as screening recommendations vary for these populations. Our growing understanding of oncogenic genetic mutations and other **breast** cancer risk factors has led to the development of several **breast** cancer risk prediction models, such as the Gail model and the Tyrer-Cuzick model. These models facilitate the stratification of **breast** cancer screening based on an individual's lifetime risk of developing the disease.[4]

In addition to mammography, ultrasonography and **breast** MRI are recommended for screening for high-risk women.[5] High-risk patients, symptomatic individuals, and **breast** cancer survivors should be identified and referred to a **breast** center for evaluation by a **breast** cancer specialist to receive appropriate screening recommendations. Please see StatPearls' companion resource, "**Breast** Cancer," for more information on high-risk factors for **breast** cancer.

Anatomy and Physiology

The mature adult **breast** comprises skin, subcutaneous tissue, and epithelial and stromal components. The epithelial component includes branching ducts that connect the structural and functional units of the **breast**, known as lobules, to the nipple. The stromal component, made up of fibrous and adipose tissue, constitutes the majority of the **breast's** volume in a non-lactating state.

Breast tissue extends vertically from the second to the sixth ribs and horizontally from the sternal edge to the midaxillary line. A portion of the **breast** tissue, known as the axillary tail of Spence, projects into the axilla. The skin of the **breast** is thin and contains sebaceous glands, exocrine sweat glands, and hair follicles. While the nipple lacks hair follicles, it is rich in sensory nerve endings and contains sebaceous and apocrine glands. The areola, which measures approximately 16 to 60 mm, is nearly circular and features increased pigmentation. Elevations near the periphery of the areola, known as Morgagni tubercles, form due to the openings of large sebaceous glands called Montgomery glands, which represent a transitional stage between sweat and mammary glands.

The **breast** is covered by superficial pectoral fascia, which extends into the superficial abdominal fascia of Camper. From beneath, the **breast** is encased in deep pectoral fascia, which covers the pectoralis major and serratus anterior muscles. These 2 fascial layers are interconnected by fibrous bands known as Cooper's suspensory ligaments, which provide natural support to the breasts. Most of the **breast's** blood supply comes from the internal mammary vessels. Sensory innervation primarily originates from the anterolateral and anteromedial branches of thoracic intercostal nerves T3 to T5. Lower fibers from the supraclavicular nerves of the cervical plexus also contribute to **breast** sensation.

Indications

The American Society of Breast Surgeons Screening Recommendations

The ASBrS guidelines recommend that all women aged 25 or older undergo a formal **breast** cancer risk assessment, with updates provided at appropriate intervals. Screening mammography should be discontinued when a woman's life expectancy is less than 10 years.

Women with an average risk of **breast** cancer and normal **breast** density should initiate yearly screening mammography, preferably 3-dimensional (3D) mammography, starting at age 40 without the need for supplemental imaging. For women with an average risk of **breast** cancer and heterogeneously or extremely dense breasts, yearly screening mammography, also preferably 3D, should start at age 40, with supplemental imaging considered.

Women at higher-than-average risk of **breast** cancer due to hereditary susceptibility from pathogenic mutation carrier status or prior chest wall radiation between the ages of 10 and 30 should begin yearly screening mammography at age 30 and be offered annual supplemental MRI starting at age 25. Additionally, women with a higher-than-average risk of **breast** cancer due to a predicted lifetime risk greater than 20% from any model or a strong family history should undergo yearly screening mammography and be offered annual supplemental MRI starting at age 35, as recommended by their clinician.

For women aged 50 or older with a prior history of **breast** cancer and non-dense breasts who did not undergo bilateral mastectomy, annual mammography, preferably 3D, is recommended. For women younger than 50 or those with dense breasts who have a prior history of **breast** cancer and did not undergo bilateral mastectomy, annual 3D mammography along with yearly supplemental MRI starting at age 35, as recommended by their clinician, is advised.

National Comprehensive Cancer Network Screening Recommendations

The 2024 NCCN Guidelines for **Breast** Cancer Screening and Diagnosis of Asymptomatic Average-Risk Patients include various recommendations (see **Table 1**. The 2024 NCCN **Breast** Cancer Screening Guidelines).[6]

A small thumbnail image of a table with multiple columns and rows, representing the 2024 NCCN Breast Cancer Screening Guidelines. The text within the table is illegible due to the small size.

Table
Table 1. The 2024 NCCN **Breast** Cancer Screening Guidelines.

Supplemental screening options such as **breast** MRI with and without contrast, abbreviated **breast** MRI, ultrasound, molecular **breast** imaging, or contrast-enhanced mammography can improve cancer detection rates but may also increase recall rates and lead to benign **breast** biopsies. For individuals at high risk of **breast** cancer, current evidence and the US Food and Drug Administration (FDA) safety announcement on gadolinium-based contrast agents support the continued recommendation for annual MRI in combination with annual screening mammography with tomosynthesis following shared decision-making.

For high-risk individuals who cannot undergo **breast** MRI, supplemental screening with contrast-enhanced mammography or molecular **breast** imaging is recommended. Whole **breast** ultrasound may be considered if contrast-enhanced or functional imaging is unavailable or inaccessible. Individuals with a residual lifetime **breast** cancer risk of 15% to 20% may be considered for supplemental screening based on individual risk factors. Abbreviated MRI demonstrates a higher cancer detection rate than mammography with tomosynthesis or ultrasound and likely offers similar sensitivity compared to a full diagnostic **breast** MRI protocol.

The NCCN panel recommends that women with average risk between the ages of 25 and 39 undergo a clinical assessment, risk-reduction counseling, and a clinical **breast** examination every 1 to 3 years. They should also be advised to notify their clinician immediately if they notice any changes in their breasts.[7]

American Cancer Society Screening Recommendations

The American Cancer Society (ACS) recommends the following **breast** cancer screening guidelines:

- Women with an average risk should undergo regular screening mammography starting at age 45.
- Women aged 45 to 54 should undergo annual screening.
- Women aged 55 and older can transition to biennial screening or continue annual screening.[8]
- Women aged 40 to 44 have the option to start annual mammography.
- Women aged 40 to 49 or older than 70 are not strongly recommended for routine screening.[9] However, in collaboration with mammography, **breast** MRI is considered an important screening modality for high-risk women and those with dense breasts.[5]
- Women who have up to 10 years of life expectancy and good health in general are advised to continue screening mammography.[10]

The United States Preventive Services Task Force Breast Cancer Screening Recommendations

The 2024 USPSTF **Breast** Cancer Screening guidelines represent the latest recommendations (see **Table 2**. The United States Preventive Services Task Force **Breast** Cancer Screening Recommendations).[3]



Table

Table 2. The United States Preventive Services Task Force **Breast** Cancer Screening Recommendations.

Notably, from 2016 to 2024, the USPSTF previously recommended biennial mammography for women aged 50 to 74. Screening was initially recommended for women aged 40 to 49 only after discussing and evaluating the risks and benefits of this test with their clinician.[7]

American College of Obstetricians and Gynecologists Screening Recommendations

ACOG released the age-based recommendations mentioned below for **breast** cancer risk assessment and screening in average-risk women, which were reaffirmed in 2021.

Patients aged 19 and older:

- Annual clinical **breast** examination.
- **Breast** self-awareness is encouraged to help detect palpable **breast** cancer; however, routine **breast** self-examinations are not recommended.

Patients aged 40 to 75:

- Screening mammography should be performed every 1 to 2 years. Women at average risk for **breast** cancer should undergo screening mammography every 1 or 2 years. This decision should be based on an informed, shared decision-making process that includes discussions about the benefits and harms of annual versus biennial screening, as well as the incorporation of patient values and preferences.
- Biennial screening mammography, especially for women aged 55 and older, is a reasonable option to minimize potential harm. However, patient counseling should

emphasize that reducing the frequency of screening may also lead to a decrease in the benefits of early detection.

- Women at average risk of **breast** cancer should continue screening mammography until at least age 75. Decisions regarding the continuation or discontinuation of screening should not be based solely on age.

Patients aged 75 and older:

- The decision to discontinue screening mammography should be guided by a shared decision-making process that takes into account the woman's health status and life expectancy.
- While **breast** self-awareness can help detect palpable **breast** cancer and is encouraged, routine **breast** self-examinations are not recommended.

World Health Organization Screening Recommendations

In 2014, the WHO Position Paper on Mammography Screening outlined the below **breast** cancer screening guidelines.

- In well-resourced settings, organized population-based mammography screening programs are recommended for women aged 50 to 69 every 2 years. For women aged 40 to 49 and those aged 70 to 75, screening programs are suggested only if conducted within a framework of rigorous research, monitoring, and evaluation.
- In limited-resource settings with relatively strong health systems, organized population-based mammography screening programs are recommended for women aged 50 to 69, conducted every 2 years, provided the conditions for implementing such a program are met. The WHO advises against mammography screening for women aged 40 to 49 and those aged 70 to 75 in these settings.
- In limited-resource settings with weak health systems, organized population-based mammography screening programs may not be cost-effective or feasible. Prioritizing early diagnosis and prompt treatment for symptomatic women is essential. Clinical **breast** examinations appear to be a promising screening approach in these settings.[11]

Contraindications

The screening guidelines from various professional societies do not uniformly apply to the patient groups mentioned below.

- Patients with short life expectancy: This includes individuals with a minimal life expectancy due to other conditions (eg, hospice patients) for whom an asymptomatic **breast** cancer diagnosis would not significantly affect their expected lifespan.[9]
- Symptomatic patients: This includes individuals who present with new palpable **breast** masses, skin changes, or nipple discharge, requiring prompt diagnostic workup. These patients should be evaluated at a **breast** center by a **breast** cancer specialist to determine the appropriate treatment.
- **Breast** cancer survivors: A patient's **breast** cancer history and prior treatment may alter screening recommendations. For instance, individuals who have had a total mastectomy typically do not require screening imaging. However, patients with a history of mammographically occult cancer who have undergone a partial mastectomy may need MRI screening in addition to routine mammography. These patients should be evaluated by a **breast** cancer specialist at a dedicated **breast** center to determine the most appropriate screening strategy.

- High-risk patients: Individuals with a greater than 20% lifetime risk of developing **breast** cancer based on any risk assessment model, which considers factors such as family history, genetics, **breast** density, and personal history, are classified as high-risk. This group also includes those with biopsy-proven high-risk lesions. These patients should be evaluated at a dedicated **breast** center by a **breast** cancer specialist to determine the most appropriate screening approach.

Equipment

Breast Mammography

Mammography is a low-dose x-ray modality used for detailed imaging of the **breast** and is considered the most effective population-based screening method. This technique can detect microcalcifications smaller than 100 μm , allowing for the identification of lesions before they become palpable. Mammography is performed in 2 forms—screening and diagnostic.

A screening mammogram is the appropriate study for asymptomatic patients at average risk. A diagnostic mammogram is required for further evaluation if an abnormality is detected during screening.[12] Tomosynthesis and 3D mammography are preferred methods due to their enhanced ability to detect small cancers and reduce the likelihood of false positives.

Breast Imaging Reporting and Data Systems (BI-RADS) are used to guide **breast** cancer diagnostic evaluation. BI-RADS categorizes **breast** lesions in a standardized format, allowing for consistent interpretation among radiologists. However, most screening mammograms do not reveal evidence of cancer upon further testing. Approximately 1% to 2% of screening mammograms identify an abnormality that necessitates a biopsy, and around 80% of the biopsied lesions are benign.

Breast Imaging Reporting and Data Systems Scoring

BI-RADS stratifies imaging findings into the categories below and provides recommendations for further diagnostic evaluation based on these classifications.

- BI-RADS 0: Additional information is needed, and further imaging may be required. Prior mammograms may also be necessary for comparison. Specific recommendations for follow-up imaging should be outlined in the radiology report.
- BI-RADS 1: No abnormalities were detected on imaging. Routine screening should be continued.
- BI-RADS 2: Benign findings observed on imaging. Continued routine screening is recommended.
- BI-RADS 3: Findings are probably benign; the growth is unlikely to be cancerous but should be reimaged to confirm stability. A repeat mammogram is recommended within the next 6 months.
- BI-RADS 4: Findings are suspicious for cancer and may necessitate a biopsy.
- BI-RADS 5: Imaging findings are highly suggestive of cancer; the growth is likely malignant and may require a biopsy.
- BI-RADS 6: Known biopsy-proven malignancy, cancer is confirmed, and treatment should begin immediately.[13]

Additional Breast Imaging Studies

MRI of the **breast** is generally more expensive than mammography, and it requires intravenous gadolinium contrast and takes longer to perform. However, MRI is more sensitive than mammography for detecting specific **breast** lesions and provides expanded views of the lymph

nodes. In high-risk women, screening MRI is considered less specific but more sensitive than mammography for detecting invasive cancers.[5] **Breast** MRI is not recommended for women at average risk; however, it is advised for those with an elevated lifetime risk of **breast** cancer. This includes women with dense **breast** tissue, some patients with silicone **breast** implants, and individuals who cannot tolerate the discomfort caused by traditional mammography compression.

Thermography techniques measure elevated skin temperatures overlying **breast** cancers but are not routinely used in screening or included in standard screening guidelines.[14] Ultrasound is usually used to evaluate abnormalities detected during a clinical examination or screening mammography. This imaging modality can aid in assessing patients with dense breasts as an adjunct to screening mammograms. However, ultrasound has limited use as a primary screening technique due to various factors, including the inability to find microcalcifications and poor specificity.

Personnel

Patients may sometimes detect **breast** cancer by noticing changes in their breasts. They should be advised to inform their clinician of any changes occurring in their breasts. Any alteration in the **breast** should prompt a clinical assessment, with further diagnostic evaluation performed as indicated by current guidelines.

Primary clinicians should remain vigilant for patients at high risk for **breast** cancer, defined as having a greater than 20% lifetime risk according to any **breast** cancer risk assessment model (eg, Gail and Tyrer-Cuzick). For high-risk screening recommendations, these patients should be evaluated by a **breast** cancer specialist at a dedicated **breast** cancer center. Additionally, patients with a strong family history of cancers—particularly **breast**, ovarian, prostate, stomach, or pancreatic cancer—should be considered for genetic testing and referred to a genetic counselor if they express interest.

Preparation

Periodic assessment of the patient's personal and family history is essential for healthcare clinicians. This evaluation should include risk factors, prior biopsy results, radiation exposure, and family history of **breast** cancer. Identifying women who may benefit from genetic counseling and high-risk screening by a **breast** cancer specialist is crucial.[12]

Recent studies promote the concept of **breast** self-awareness, which involves being attuned to the appearance and feel of a woman's breasts and the ability to notice any changes. Women are encouraged to report any changes to their primary healthcare clinician. In contrast, **breast** self-examination involves a regular and systematic approach to assessing the breasts.[15]

Technique or Treatment

Imaging techniques for **breast** cancer screening are most effective when considering both sensitivity and specificity while also accounting for potential complications and harms to the screening population. The preferred imaging method is mammography, followed by commonly used modalities such as ultrasonography and MRI. Additionally, **breast** self-examination and clinical **breast** examinations remain important components of **breast** cancer detection.

Screening methods, such as mammography, are most effective when targeted strategies are used, considering age and other factors, including hormonal exposure, family history, and risk factors such as radiation, obesity, and genetics. MRI involves the injection of intravenous contrast material, which enhances the ability to differentiate normal **breast** tissue from abnormal lesions.

Complications

Mammography has its limitations as a screening imaging method. Research indicates that it is less sensitive in detecting cancer in dense **breast** tissue, which can result in false-negative outcomes and the potential for missed cancers, if present. Some **breast** cancers may be mammographically occult, failing to appear on traditional mammograms. Thus, it is essential to conduct imaging alongside a physical examination by a healthcare clinician. Further diagnostic imaging, such as MRI or ultrasound, should be ordered when discrepancies arise. Additionally, high-risk patients should be screened appropriately, often utilizing MRI for more accurate detection.

Increased treatment interventions are associated with screening, which may not always be effective or necessary. Individuals who undergo screening are more likely to experience surgery and radiation therapy. Overscreening can have negative consequences for patients, including economic, psychological, cosmetic, physical, and productivity-related harm. The overall lifetime radiation exposure increases with the age of initiation, frequency, and cessation of screening, as women are exposed to approximately 3.7 mGy per digital mammogram. This raises the risk of radiation-induced **breast** cancer to 125 cases per 100,000 women aged 40 to 74, potentially leading to an increase in deaths attributed to **breast** cancer screening.

Ultrasonography is generally considered a highly operator-dependent modality and supplemental screening test that requires a skilled practitioner, high-quality examination, and state-of-the-art equipment. Based on the findings from various studies, a prospective, multicenter study is warranted to further investigate the role of this imaging modality in **breast** cancer screening.[16]

Clinical Significance

Age should not be the sole factor in determining **breast** cancer screening. A comprehensive assessment of all relevant risk factors—including **breast** density, genetics, and personal history—should guide the screening process.[17] Additionally, the sensitivity and specificity of mammography tend to improve with advancing age compared to younger women.

Dense **breast** tissue is a common finding on mammograms, particularly in younger women, as **breast** density typically decreases with age. However, dense breasts, especially in older patients, are associated with a higher likelihood of developing **breast** cancer, and small lesions can be more challenging to detect on traditional mammograms. While mammography may reduce sensitivity in detecting **breast** cancer among women with dense breasts, alternative screening strategies such as MRI and ultrasonography can be utilized. These methods are particularly recommended for women at higher risk of developing **breast** cancer.

An early age of initiation for screening and the use of MRI and ultrasound may be considered for women with first-degree relatives diagnosed with **breast** cancer.[5] For BRCA mutation carriers, a combination of annual **breast** MRI and mammography is recommended for **breast** cancer screening. It is essential to accurately identify patients with a greater than 20% lifetime risk of **breast** cancer and to recognize that standard screening guidelines for asymptomatic average-risk patients do not apply to this population. High-risk individuals should be referred to a **breast** cancer specialist for tailored screening recommendations and, if appropriate, to a genetic counselor for genetic testing.

Enhancing Healthcare Team Outcomes

Screening average-risk **breast** cancer patients necessitates a collaborative and interprofessional approach to enhance patient-centered care, improve outcomes, and ensure safety. Physicians, advanced practitioners, nurses, pharmacists, radiologists, and technicians must work together to balance the benefits and risks of screening. Clinicians are responsible for identifying risk factors and providing appropriate screening based on age, **breast** density, and family history. Effective counseling, especially for patients with genetic predispositions, requires shared decision-making. Radiologists and technicians must skillfully perform and interpret imaging to minimize false

positives and negatives. This coordinated effort fosters a comprehensive care strategy that improves diagnosis and patient outcomes.

Review Questions

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